

Clustering: Unsupervised Learning

IF-3170 Inteligensi Artifisial
Teknik Informatika ITB

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01 Clustering: What & Why?

Outline

Clustering: What?

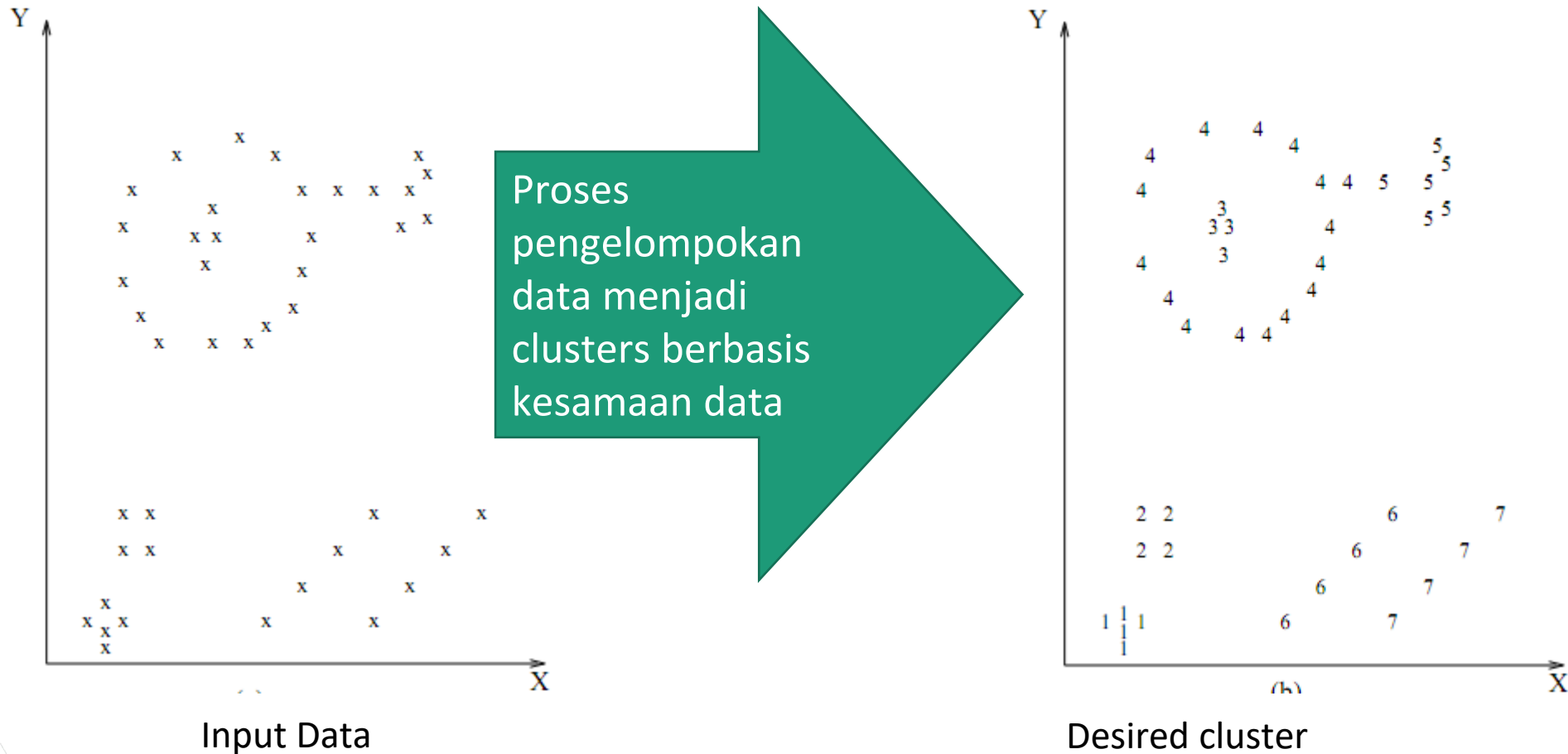
Tahapan Clustering

Representasi
Cluster

Clustering: Why?

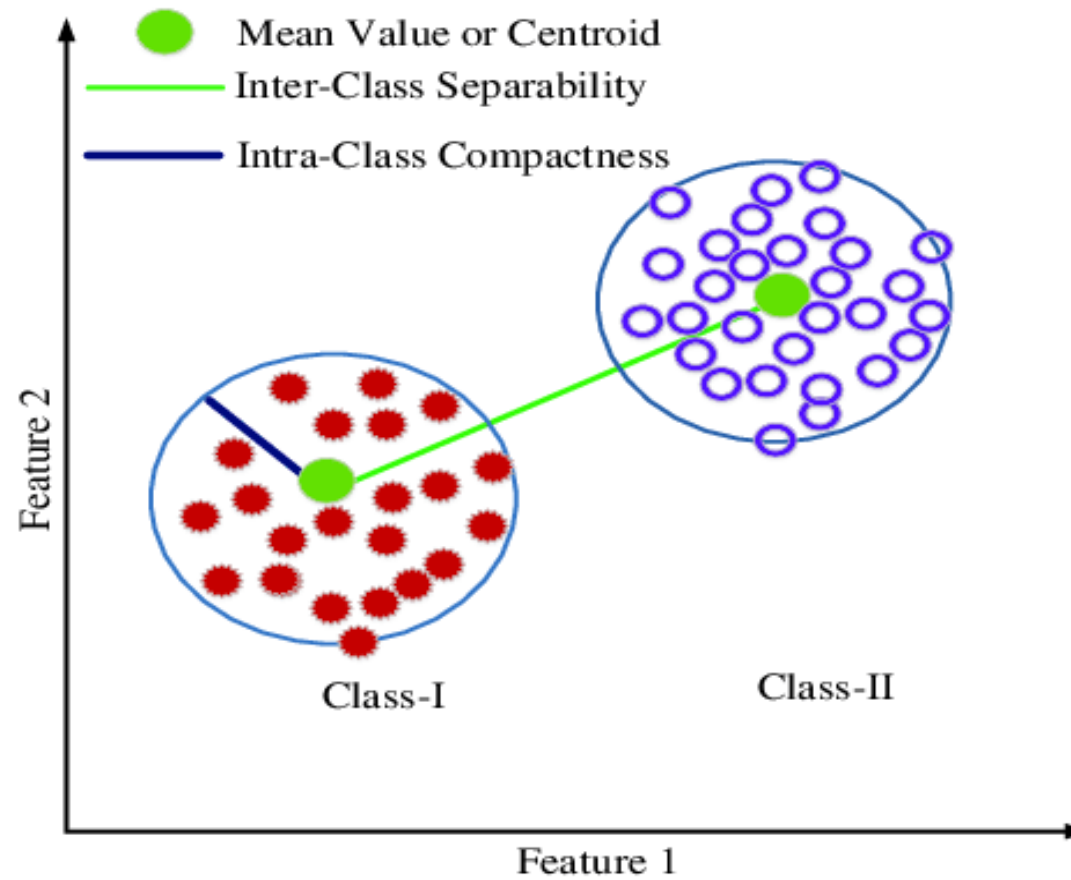
Kategori Metode

Clustering: What ?



Unsupervised learning = learning from raw data

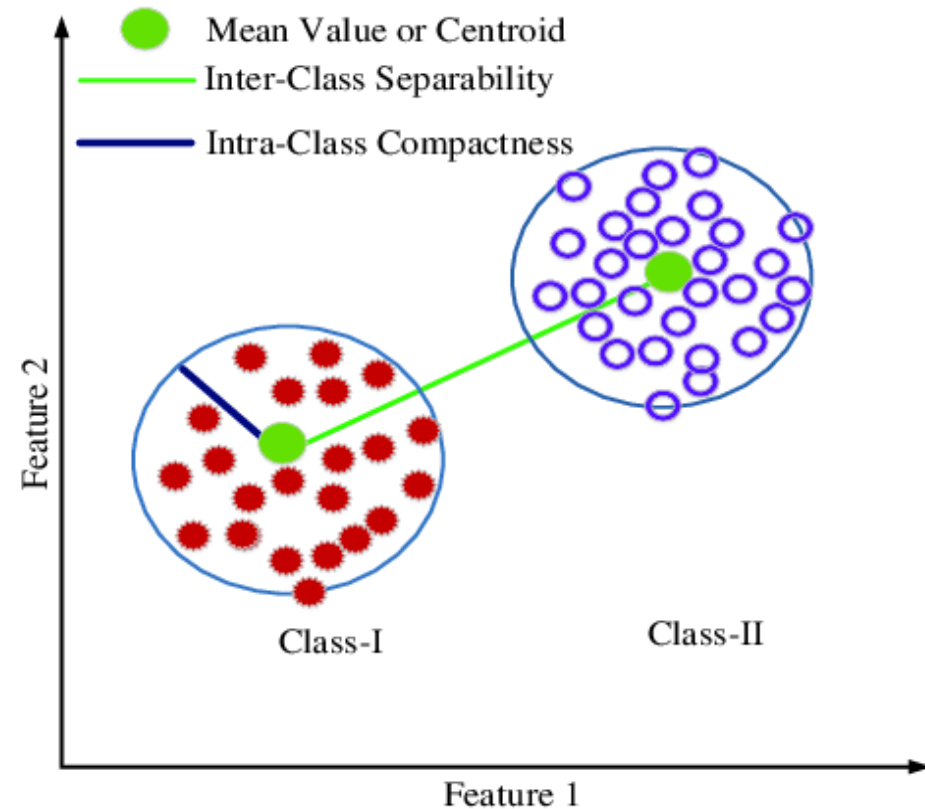
Clustering:



https://www.researchgate.net/figure/The-concept-of-intra-class-compactness-and-inter-class-separability-in-a-two-dimension_fig3_325095062

Clustering: Finding natural groups

- High intra-cluster similarity
 - Data pd cluster yang sama harus semirip mungkin
- Low inter-cluster similarity
 - Data pd cluster yang berbeda harus sejauh mungkin
- Pengukuran kemiripan dan jarak harus jelas dan punya semantik praktikal (sesuai domain)

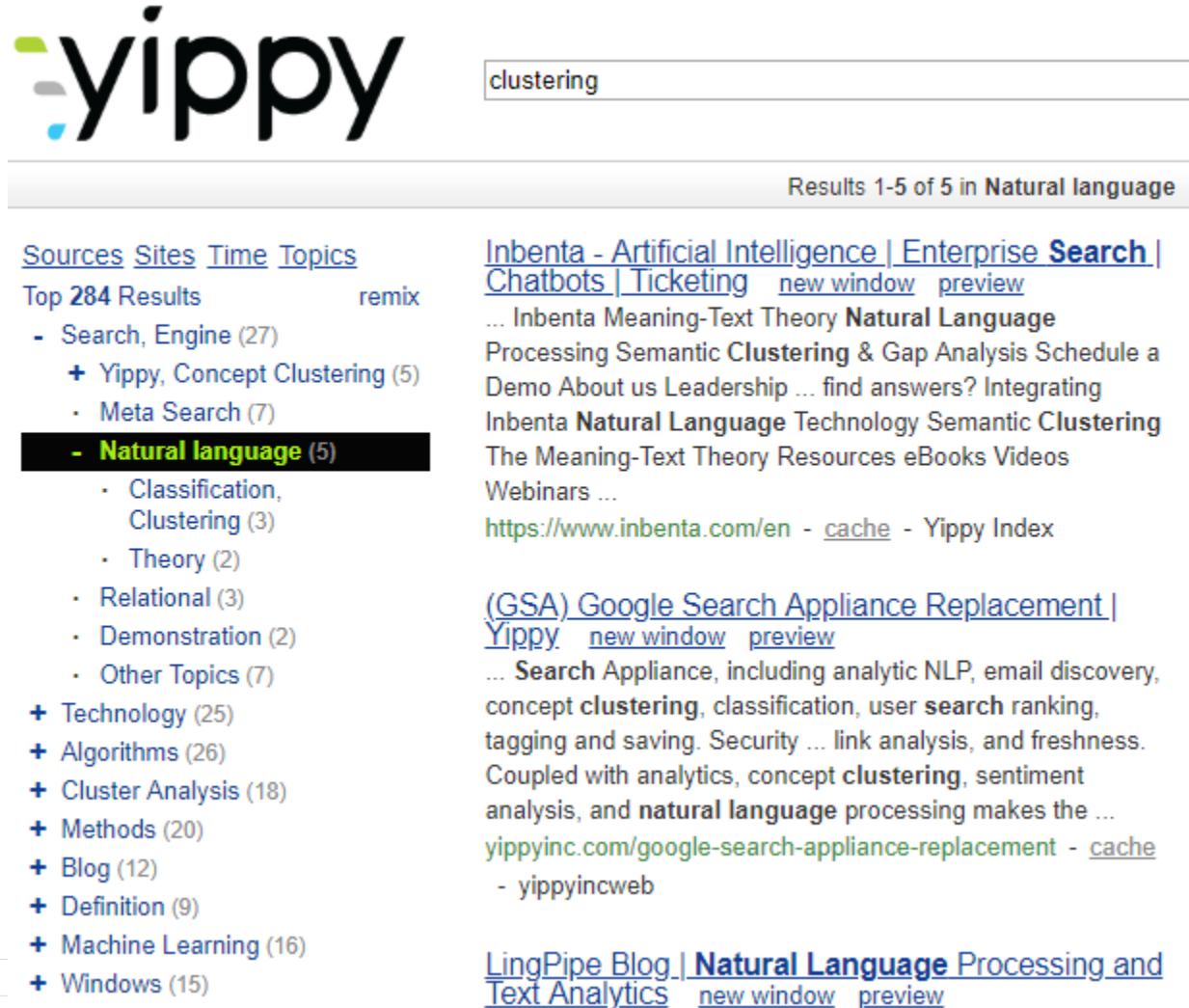


https://www.researchgate.net/figure/The-concept-of-intra-class-compactness-and-inter-class-separability-in-a-two-dimension_fig3_325095062

Clustering: Why ?

- Data discovery (cluster = struktur internal data)
 - Contoh: search engine, news aggregator, gen
- Tujuan awalnya partisi / pengelompokan
 - Contoh: segmentasi pasar, segmentasi gambar
- Bagian dari teknik lainnya
 - Contoh: peringkasan berbasis clustering

Why: Clustering pada Search Engine



yippy

Results 1-5 of 5 in Natural language

[Sources](#) [Sites](#) [Time](#) [Topics](#)

Top 284 Results remix

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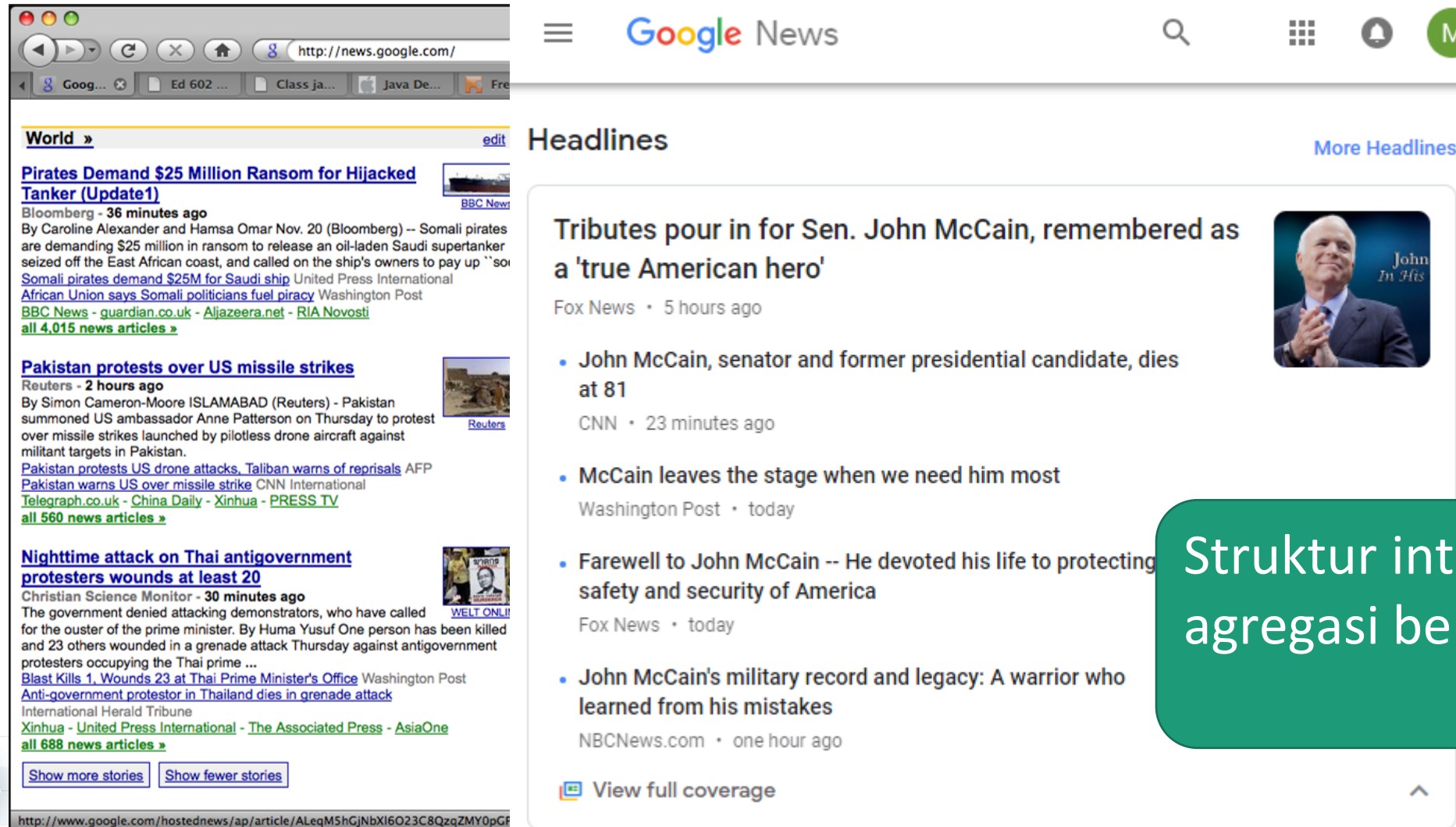
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Struktur internal
hasil pencarian



Why: Clustering pada News Aggregator



The screenshot shows the Google News homepage. The main section displays headlines under the 'Headlines' tab. A prominent headline reads 'Tributes pour in for Sen. John McCain, remembered as a 'true American hero'' from Fox News, 5 hours ago. Below it, other headlines include 'John McCain, senator and former presidential candidate, dies at 81' from CNN, 23 minutes ago; 'McCain leaves the stage when we need him most' from Washington Post, today; 'Farewell to John McCain -- He devoted his life to protecting safety and security of America' from Fox News, today; and 'John McCain's military record and legacy: A warrior who learned from his mistakes' from NBCNews.com, one hour ago. A 'View full coverage' link is at the bottom of the main section. On the left sidebar, under the 'World' tab, there are three news items: 'Pirates Demand \$25 Million Ransom for Hijacked Tanker (Update1)' from Bloomberg, 36 minutes ago; 'Pakistan protests over US missile strikes' from Reuters, 2 hours ago; and 'Nighttime attack on Thai antigovernment protesters wounds at least 20' from Christian Science Monitor, 30 minutes ago. Each item includes a brief summary and links to more articles. A 'Show more stories' button is at the bottom of the sidebar.

Headlines

Tributes pour in for Sen. John McCain, remembered as a 'true American hero'
Fox News • 5 hours ago

- John McCain, senator and former presidential candidate, dies at 81
CNN • 23 minutes ago
- McCain leaves the stage when we need him most
Washington Post • today
- Farewell to John McCain -- He devoted his life to protecting safety and security of America
Fox News • today
- John McCain's military record and legacy: A warrior who learned from his mistakes
NBCNews.com • one hour ago

[View full coverage](#)

World »

Pirates Demand \$25 Million Ransom for Hijacked Tanker (Update1)
Bloomberg - 36 minutes ago
By Caroline Alexander and Hamsa Omar Nov. 20 (Bloomberg) -- Somali pirates are demanding \$25 million in ransom to release an oil-laden Saudi supertanker seized off the East African coast, and called on the ship's owners to pay up -- so
[Somali pirates demand \\$25M for Saudi ship](#) United Press International
[African Union says Somali politicians fuel piracy](#) Washington Post
[BBC News](#) - [guardian.co.uk](#) - [Aljazeera.net](#) - [RIA Novosti](#)
[all 4,015 news articles »](#)

Pakistan protests over US missile strikes
Reuters - 2 hours ago
By Simon Cameron-Moore ISLAMABAD (Reuters) - Pakistan summoned US ambassador Anne Patterson on Thursday to protest over missile strikes launched by pilotless drone aircraft against militant targets in Pakistan.
[Pakistan protests US drone attacks, Taliban warns of reprisals](#) AFP
[Pakistan warns US over missile strike](#) CNN International
[Telegraph.co.uk](#) - [China Daily](#) - [Xinhua](#) - [PRESS TV](#)
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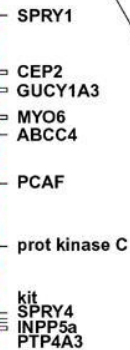
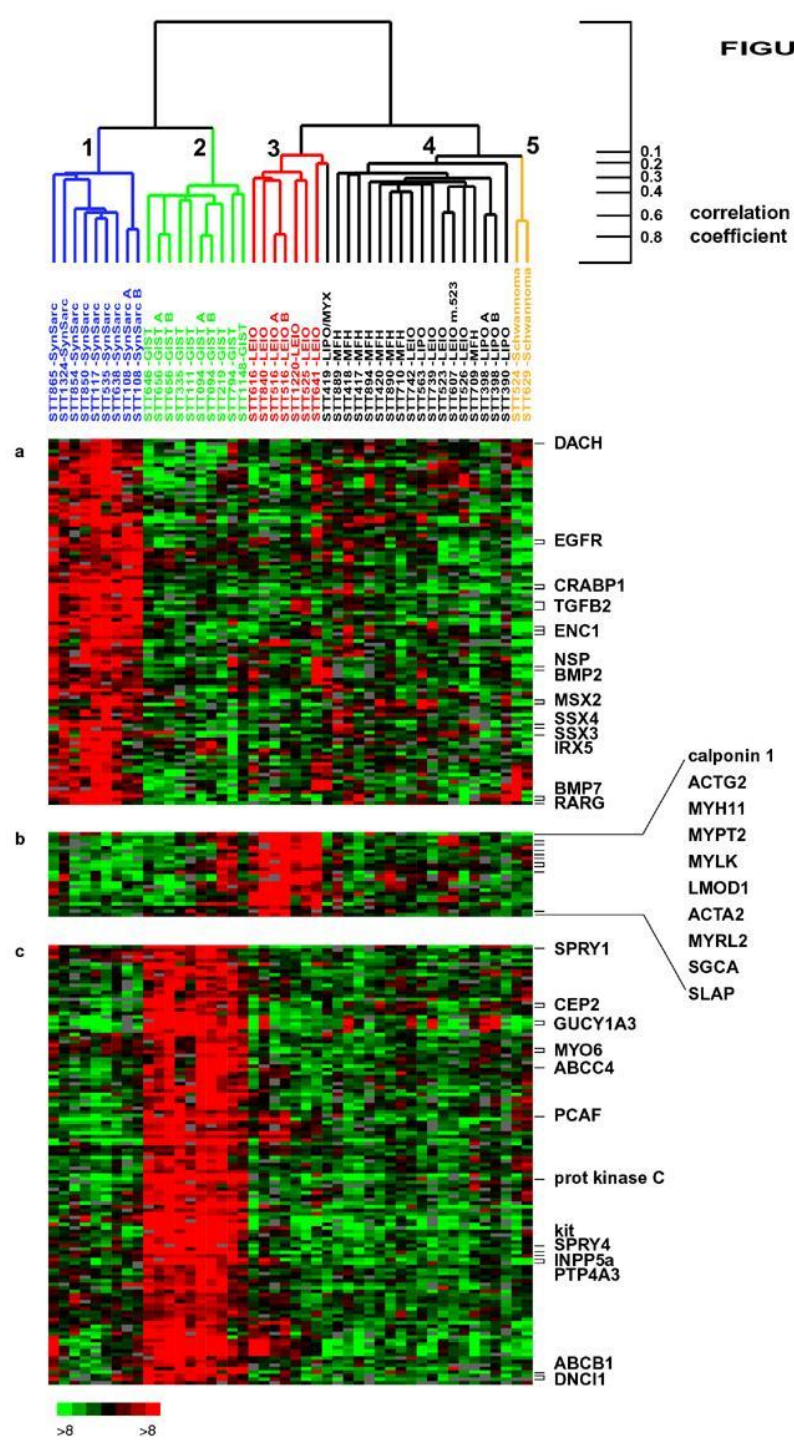
Nighttime attack on Thai antigovernment protesters wounds at least 20
Christian Science Monitor - 30 minutes ago
The government denied attacking demonstrators, who have called for the ouster of the prime minister. By Huma Yusuf One person has been killed and 23 others wounded in a grenade attack Thursday against antigovernment protesters occupying the Thai prime ...
[Blast Kills 1, Wounds 23 at Thai Prime Minister's Office](#) Washington Post
[Anti-government protestor in Thailand dies in grenade attack](#) International Herald Tribune
[Xinhua](#) - [United Press International](#) - [The Associated Press](#) - [AsiaOne](#)
[all 688 news articles »](#)

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Struktur internal agregasi berita

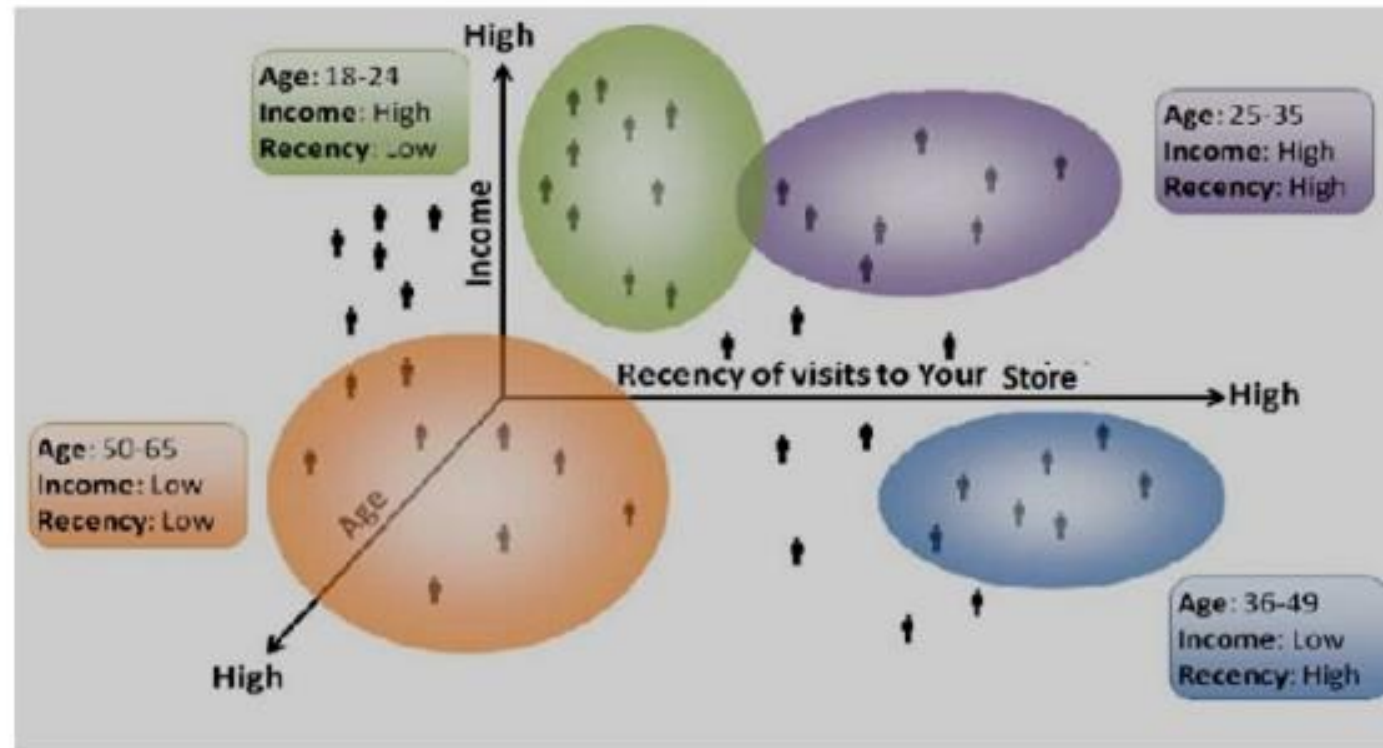


http://genome-www.stanford.edu/sarcoma/supplemental_data.html



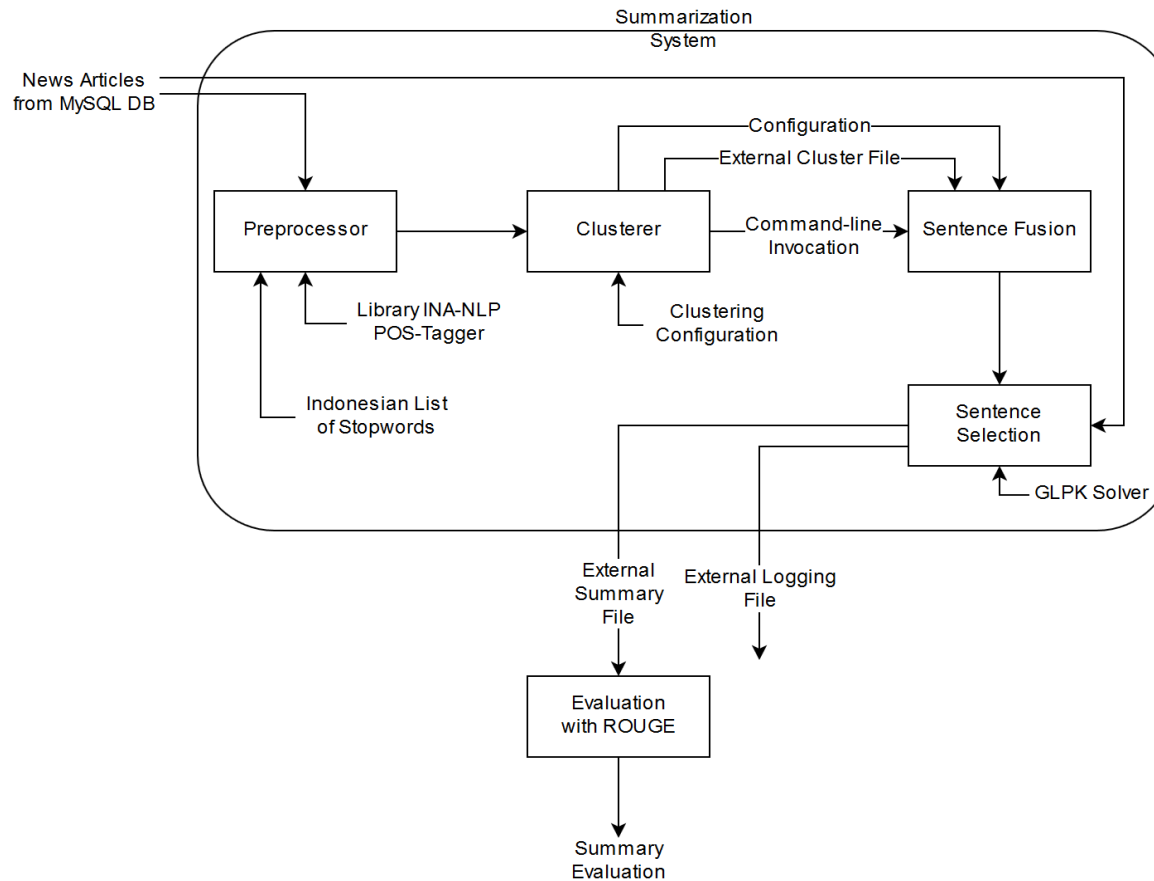
Why: Clustering untuk Segmentasi

Example - Clusters using Age, Income & Recency



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Why: Clustering-based Approach

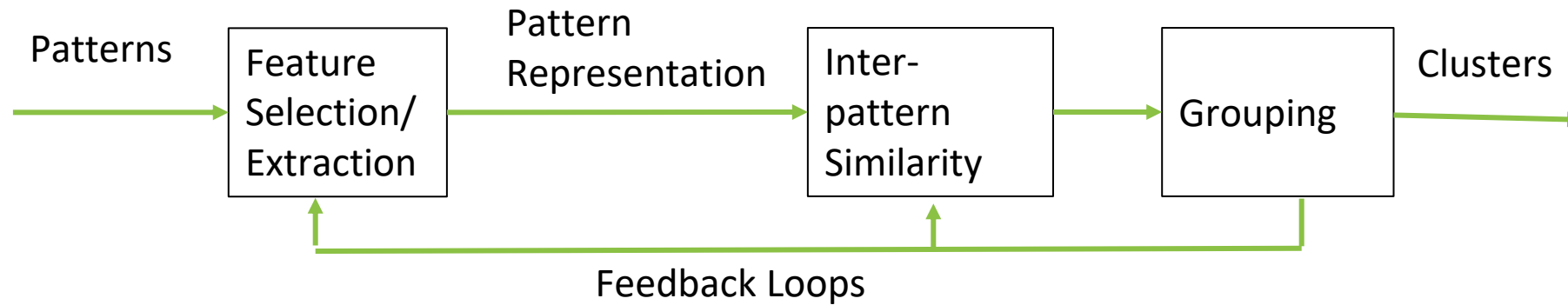


Clustering-based
summarization

Clustering-based
outlier detection

Clustering-based
analysis

Tahapan Clustering



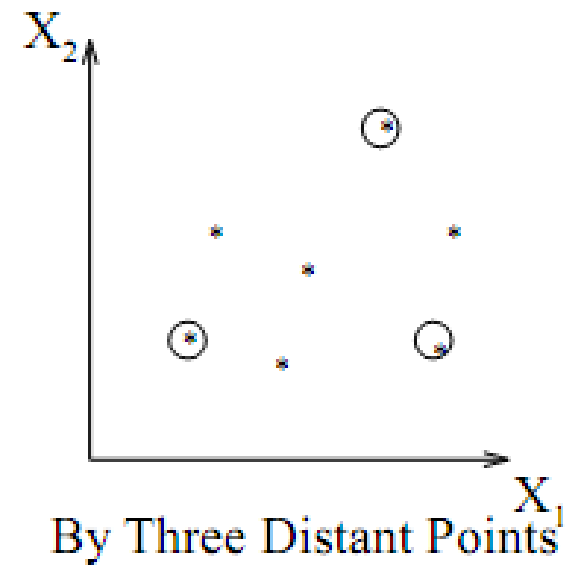
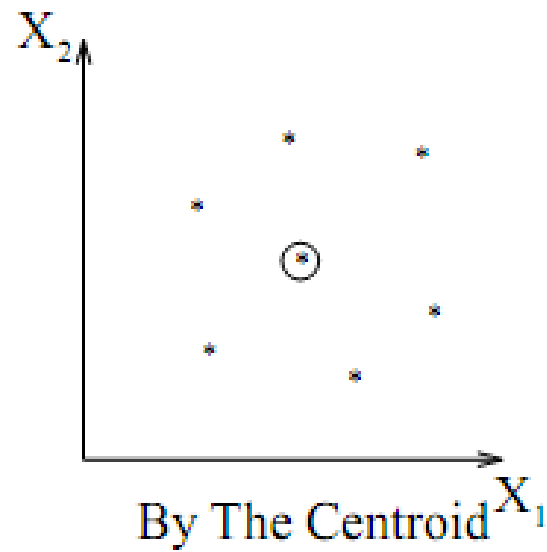
- Tahapan utama:
 - 1) Feature selection: original features $\square$ subset of features
Feature extraction: transformation into new features
 - 2) Pattern proximity/similarity measure
 - 3) Grouping
- Clustering output: hard atau soft (membership degree)

Tahapan Clustering (lanjutan)

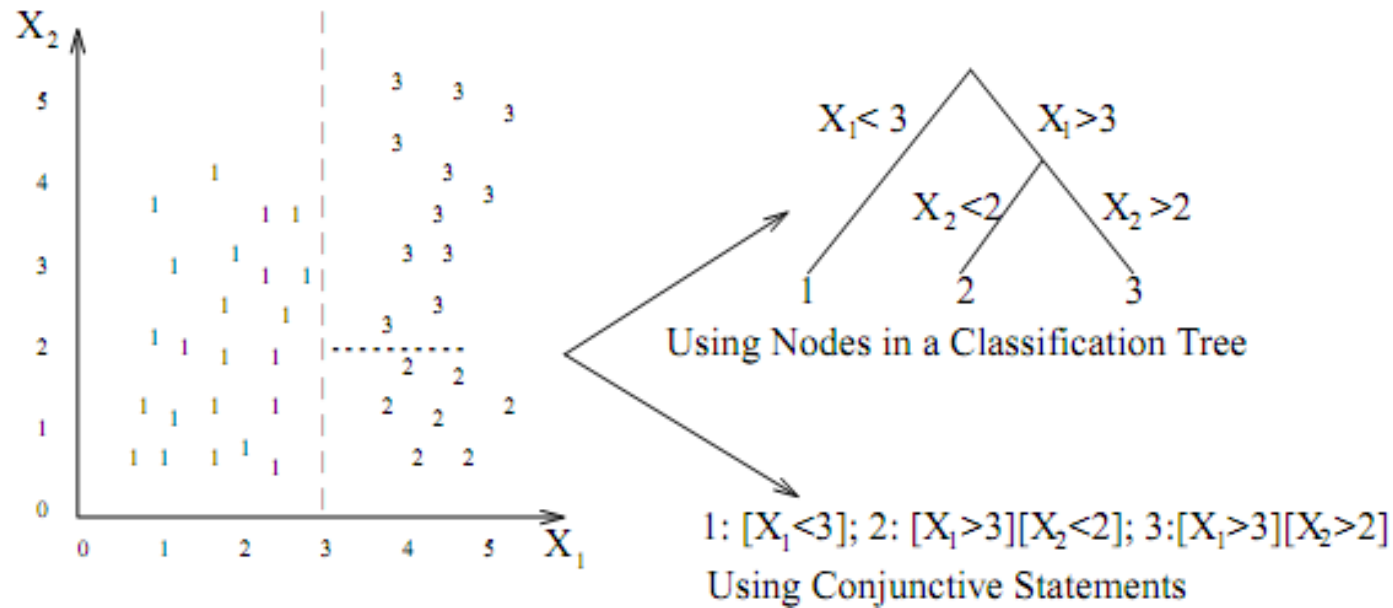
- Tahapan opsional:
 - 4) data abstraction
 - 5) assessment of output (good or poor)

Representasi Cluster (1)

- Centroid atau set of distant point

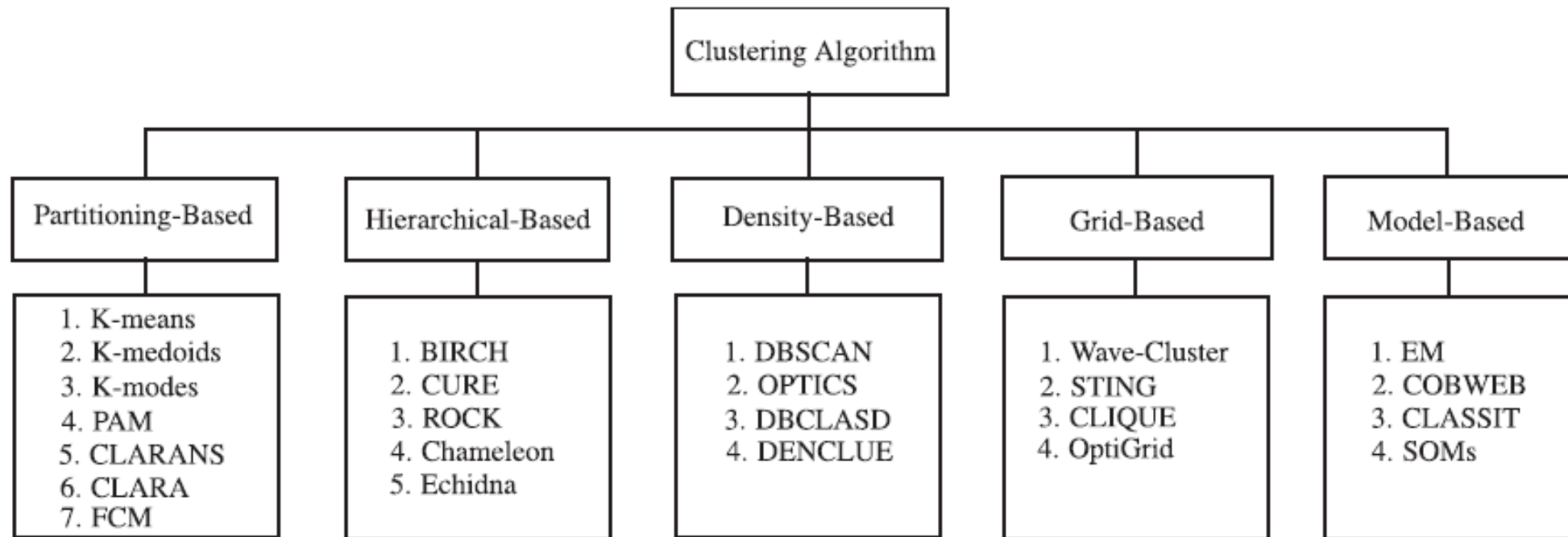


Representasi Cluster (2)



- Pohon klasifikasi
- Conjunctive statements

Kategori Metode Clustering (Fahad, 2014)



Kategori Metode Clustering

(Han & Kamber, 2006)

1. Metode *partitioning*

- mengidentifikasi partisi yang mengoptimalkan kriteria pengelompokan (squared error, absolute error)
- Konstruksi k-partisi data (partisi ~ cluster); $k \leq$ jumlah data
- Contoh: **K-means**, k-medoids

2. Metode *hierarchical*

menghasilkan rangkaian partisi bersarang

- **Agglomerative** (bottom-up, merge):
1 object ~ 1 cluster $\square$ 1 cluster n-object
- Divisive (top-down, split):
1 cluster n-object $\square$ 1 object ~ 1 cluster

Kategori Metode Clustering (lanj)

(Han & Kamber, 2006)

3. Metode berbasis density

- Densitas: jumlah objek
- Contoh: **DBSCAN** (Density-Based Spatial Clustering of Applications with Noise)

4. Metode berbasis grid

- Struktur grid, cepat, bergantung jumlah sel, tidak dipengaruhi jumlah objek, perhitungan bisa dilakukan secara paralel
- Contoh: STING (STatistical INformation Grid)

5. Metode berbasis model

- Contoh: EM (Expectation-Maximization), SOM (self-organizing map)



02 Partitional Clustering

02 Partitional Clustering K-Means

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Metode *Partitioning*

- Kriteria objektif: objek di dalam suatu cluster memiliki kemiripan yang lebih besar dibanding objek yang berada di cluster yang lain.
 - Minimize square-error function.
- Relokasi iteratif:
 - proses iteratif menempatkan objek ke kluster untuk memperbaiki partisi.
- Contoh: Squared-Error Clustering (**k-Means**), k-medoids, Graph theoretic

k-Means

Popular clustering
algorithm

1.
Initialization

Define k seeds (as centroid) for k cluster, randomly

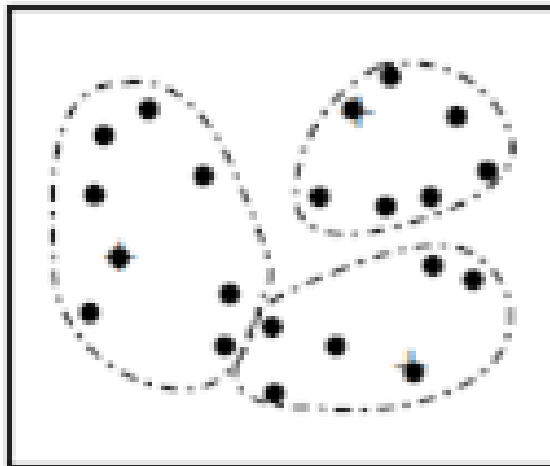
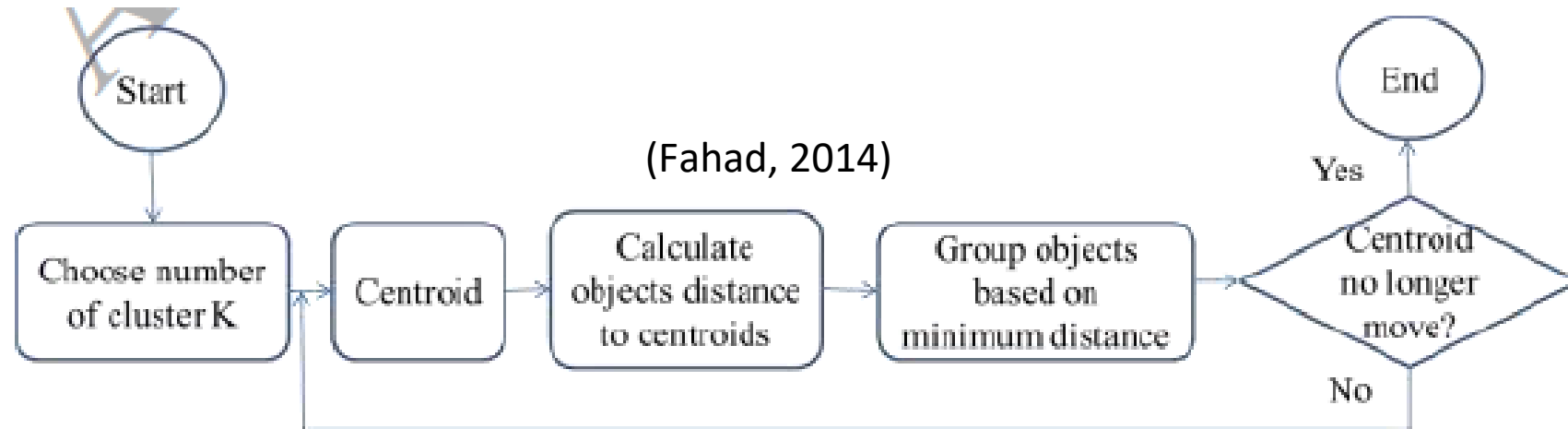
2. Iteration

Assign every instance from training data to the closest centroid, form k cluster

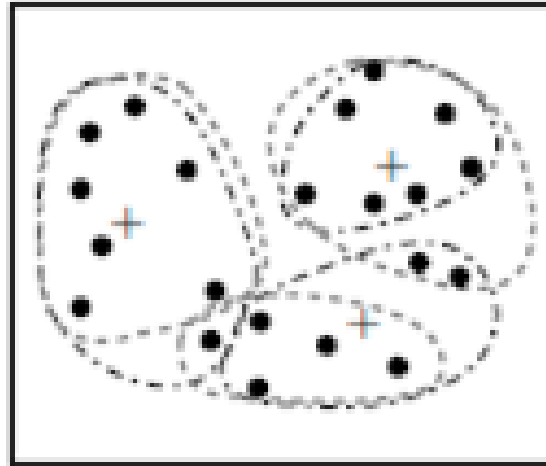
Specify the new centroid (M_i) of each cluster

Stop: Converge/ threshold/ number of iteration

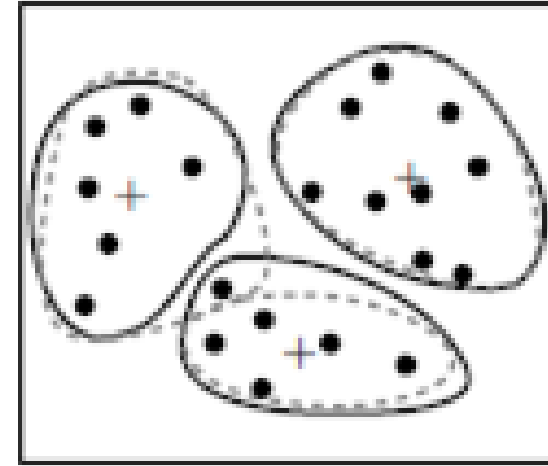
K-Means



(a)



(b)



(c)

Algoritma K-Means

(Han & Kamber, 2006)

- The k-means algorithm for partitioning, where each cluster's center is represented by mean value of the objects in the cluster.
- Input: **k** (number of clusters), D (data set containing **n** objects)
- Output: A set of k clusters
- Kompleksitas: **$O(nkt)$** ; t: jumlah iterasi; $k \ll n$
- Method:
 1. Arbitrarily choose k objects from D as the initial cluster centers
 2. Repeat
 - (re) assign each object to the cluster to which the object is the most similar based on the mean value of the objects in the clusters;
 - Update the cluster means, i.e. calculate the mean value of the objects for each cluster;
 3. Until no change (convergent)

Example

Data	Pengalaman	Programming	Bahasa Inggris	Buta Warna	Menikah
A	Tidak	Ya	Ya	Tidak	Tidak
B	Tidak	Tidak	Ya	Tidak	Ya
C	Ya	Ya	Tidak	Ya	Tidak
D	Tidak	Tidak	Tidak	Ya	Ya
E	Ya	Tidak	Ya	Ya	Ya
F	Tidak	Ya	Tidak	Ya	Tidak

$K = 3$

Seed: A,
B, C

Distance: difference of attribute value, if it has similar distance, group the instance into cluster 2

Centroid (mean) : majority value for each attribute, if no majority choose previous value

Stop:
max distance in each cluster < 2

k-Means Implementation

Initialization:

Centroid Cluster 1: Data A

Centroid Cluster 2: Data B

Centroid Cluster 3: Data C

Iteration 1:

Assign each instance to closest centroid:

Cluster 1: A

Cluster 2: B, D, E

Cluster 3: C, F

Data	Pengalaman	Programming	Bahasa Inggris	Buta Warna	Menikah	Distance Centroid 1	Distance Centroid 2	Distance Centroid 3
A	Tidak	Ya	Ya	Tidak	Tidak	0	2	3
B	Tidak	Tidak	Ya	Tidak	Ya	2	0	5
C	Ya	Ya	Tidak	Ya	Tidak	3	5	0
D	Tidak	Tidak	Tidak	Ya	Ya	4	2	3
E	Ya	Tidak	Ya	Ya	Ya	4	2	3
F	Tidak	Ya	Tidak	Ya	Tidak	2	4	1

k-Means Implementation

Initialization:

Centroid Cluster 1: Data A

Centroid Cluster 2: Data B

Centroid Cluster 3: Data C

Iteration 1:

Adjust new centroid

Centroid Cluster 1:

{Tidak, Ya, Ya, Tidak, Tidak} □ max dist: 0

Centroid Cluster 2:

{Tidak, Tidak, Ya, Ya, Ya}

□ max dist: 1

Centroid Cluster 3:

{Ya, Ya, Tidak, Ya, Tidak}

□ max dist: 1

STOP? Yes

Data	Pengalaman	Programming	Bahasa Inggris	Buta Warna	Menikah	Distance Centroid 1	Distance Centroid 2	Distance Centroid 3
A	Tidak	Ya	Ya	Tidak	Tidak	0	2	3
B	Tidak	Tidak	Ya	Tidak	Ya	2	0	5
C	Ya	Ya	Tidak	Ya	Tidak	3	5	0
D	Tidak	Tidak	Tidak	Ya	Ya	4	2	3
E	Ya	Tidak	Ya	Ya	Ya	4	2	3
F	Tidak	Ya	Tidak	Ya	Tidak	2	4	1

Usage

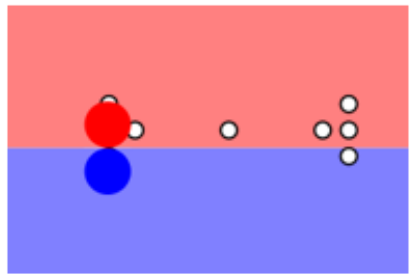
New Instance G:
{Ya, Ya, Ya, Tidak, Tidak}

Centroid Cluster 1:
{Tidak, Ya, Ya, Tidak, Tidak}
Centroid Cluster 2:
{Tidak, Tidak, Ya, Ya, Ya
Centroid Cluster 3:
{Ya, Ya, Tidak, Ya, Tidak}

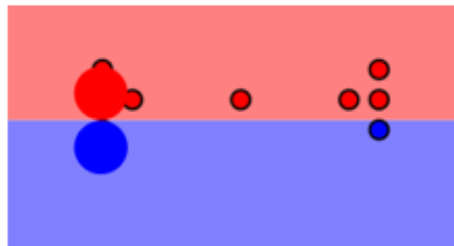
Distance Centroid 1	Distance Centroid 2	Distance Centroid 3
1	4	2

New Instance G:
Cluster 1

K-Means: Pengaruh Cluster Awal

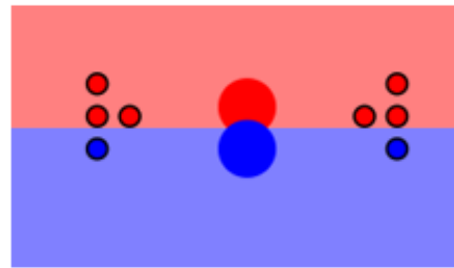


i=0

{A,B,D,E,F,G,H}, {C,I}

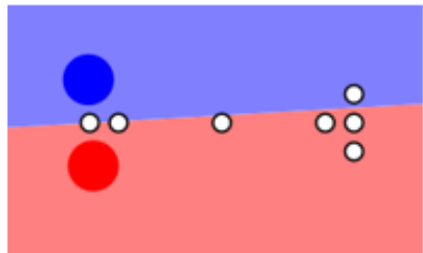
i=1

{A,B,D,E,F,G,H}, {C,I}

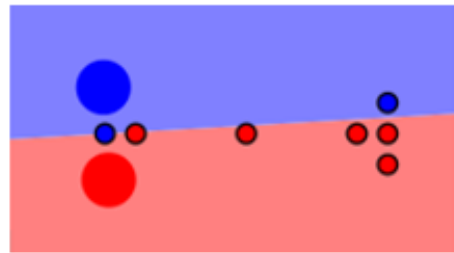


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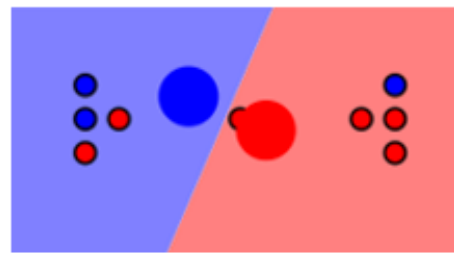


i=0

{A,G}, {B,C,D,E,F,H,I}

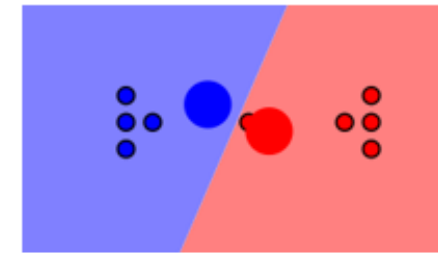
i=1

{A,B,G}, {C,D,E,F,H,I}



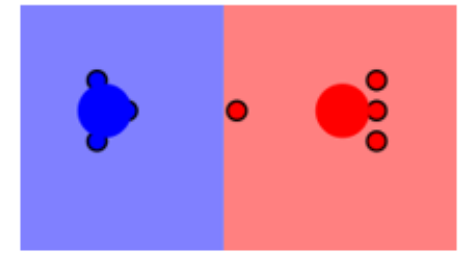
i=2

{A,B,G}, {C,D,E,F,H,I}



i=3

{A,B,C,D}, {E,F,G,H,I}



i=4

{A,B,C,D}, {E,F,G,H,I}

Kelemahan K-Means (Berkhin, 2006)

- Penentuan nilai k? ☐ Gunakan Elbow Method (slide berikutnya)
- Hasil sangat dipengaruhi oleh inisialisasi centroid
 - Sering berhenti pada optimum lokal
 - Hasil akhir tidak stabil
- Algoritma tidak *scalable*
- Mean hanya terdefinisi untuk atribut numerik
 - Atribut nominal ☐ metode k-modes
- Prosesnya sensitif terhadap outliers
 - Outlier dgn nilai yang ekstrim besar mempengaruhi mean pada centroid

Clustering with K-Means

How we know the best "K"

Within-Cluster-Sum-of-Squares: $WCSS = \sum_k^n \sum_i^m \|d_i - c_k\|^2$

Expect: Small WCSS

Elbow Method

N instances in training data,
create N cluster $\square$ WCSS =
0



Clustering with K-Means

How we know the best "K"

Within-Cluster-Sum-of-Squares:
 $WCSS = \sum_k^n \sum_i^m \|d_i - C_k\|^2$

Expect: Small WCSS

Elbow Method

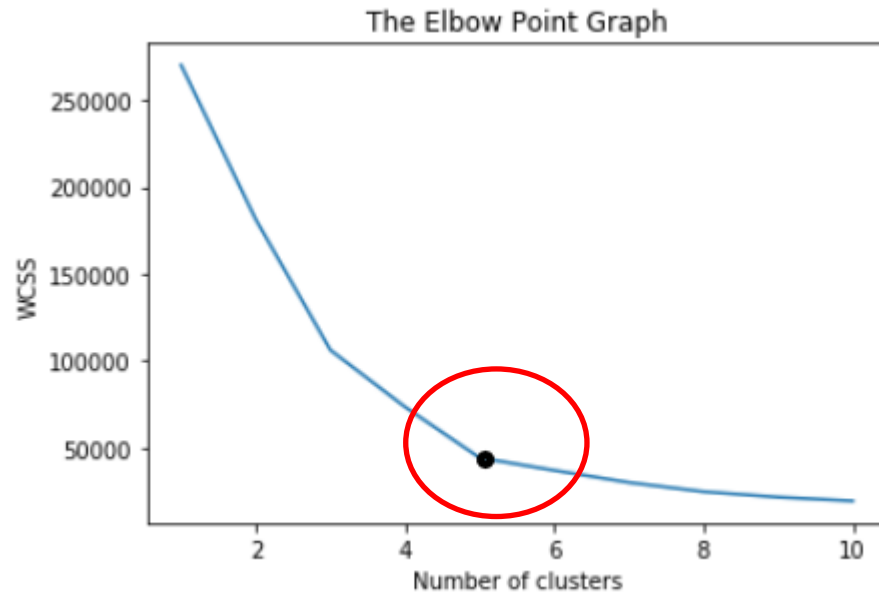
N instances in training data,
create N clusters, WCSS = 0



Elbow Method

Try k from the smallest number to the largest

For each k , run k-Means and plot its WCSS



Find k where the position is in the “elbow”

Q & A: K-means

- Q: Kompleksitas algoritma K-means $O(nkt)$, tetapi disebutkan tidak scalable.
- A: Jumlah iterasi untuk kasus terburuknya bisa mencapai eksponensial / superpolynomial.
 - Xu & Tian (2017): Kompleksitas algoritma K-means $O(nkt)$ dikategorikan kompleksitas rendah.
 - Dimensi diabaikan $\square$ $O(nktd)$; d dimensi data
 - Berkhin (2006): Scalability both in terms of computing time and memory requirements.
 - Arthur & Vassilvitskii (2006): The worst-case running time of k-means is superpolynomial number of iteration ($2^{\Omega(n)}$ iteration).





03 Density-based Clustering

03 Density-based Clustering

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Outline

DBSCAN

Density-based
Clustering

Example

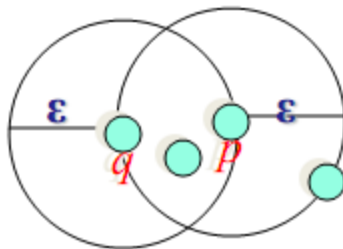
Density-based Clustering

- Clusters are dense regions in the data space, separated by regions of lower object density
- A cluster is defined as a maximal set of density-connected points

ϵ -Neighborhood – Objects within a radius of ϵ from an object.

$$N_{\epsilon}(p) : \{q \mid d(p, q) \leq \epsilon\}$$

“High density” - ϵ -Neighborhood of an object contains at least **MinPts** of objects.



ϵ -Neighborhood of p

ϵ -Neighborhood of q

Density of p is “high” (MinPts = 4)

Density of q is “low” (MinPts = 4)

DBSCAN

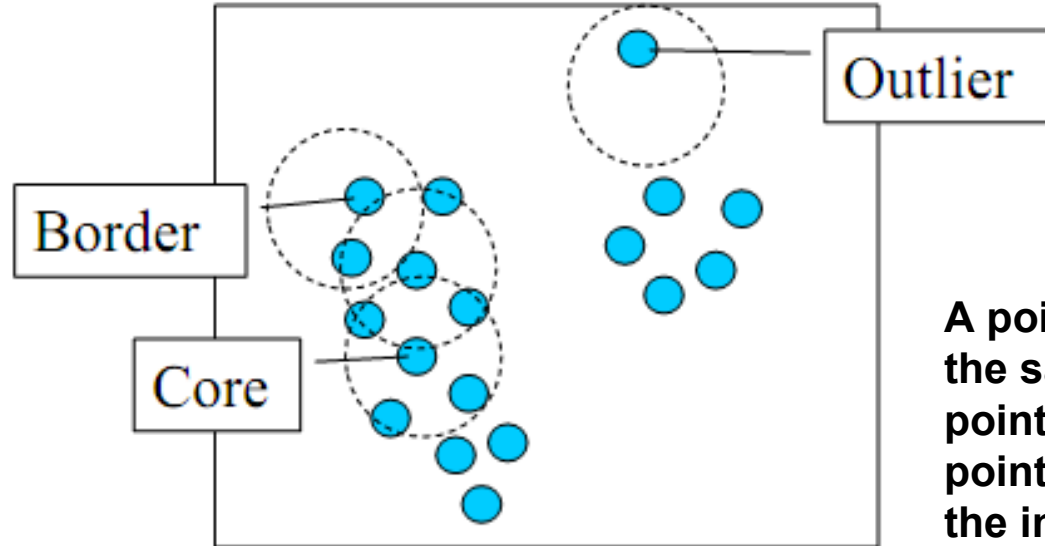
(Density Based Spatial Clustering of Applications with Noise)

- DBSCAN starts with an arbitrary point p and retrieves all points density-reachable from p wrt. Eps and $MinPts$.
 - If p is a core point, this procedure yields a cluster wrt. Eps and $MinPts$.
 - If p is a border point, no points are density-reachable from p and DBSCAN visits the next point of the database.

DBSCAN: Object Type

- The neighborhood within a radius ϵ of a given object is called the ϵ -neighborhood of the object.
- If the ϵ -neighborhood of an object contains at least a minimum number, $MinPts$, of objects, then the object is called a **core object**.
- Given a set of objects, D , we say that an object p is **directly density-reachable** from object q if p is within the ϵ -neighborhood of q , and q is a core object.
- An object p is **density-reachable** from object q with respect to ϵ and $MinPts$ in a set of objects, D , if there is a chain of objects $p_1, \dots, p_n$, where $p_1 = q$ and $p_n = p$ such that p_{i+1} is directly density-reachable from p_i with respect to ϵ and $MinPts$, for $1 \leq i \leq n$, $p_i \in D$.
- An object p is **density-connected** to object q with respect to ϵ and $MinPts$ in a set of objects, D , if there is an object $o \in D$ such that both p and q are density-reachable from o with respect to ϵ and $MinPts$.

Core, Border, Outlier



Given ε and *MinPts*, categorize the objects into three exclusive groups.

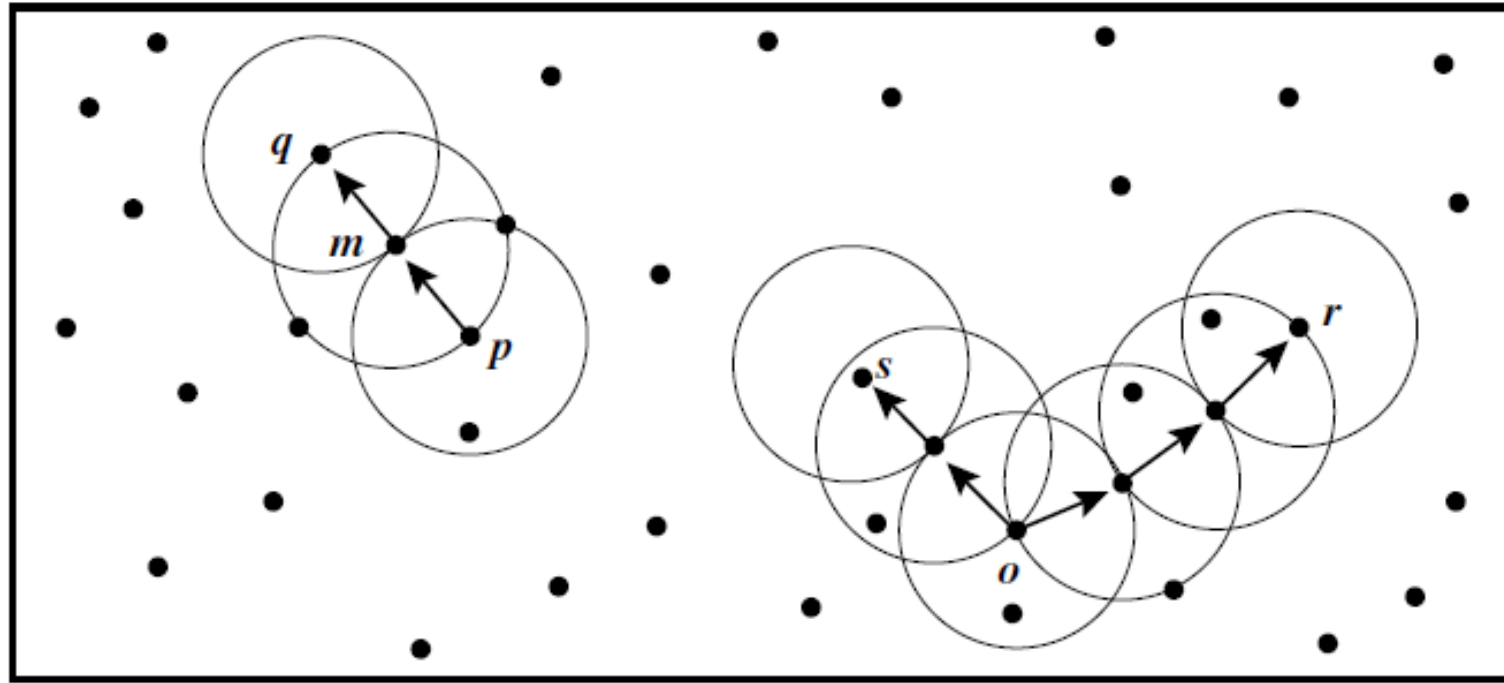
A point is a **core point** if it has at least the same as a specified number of points (*MinPts*) within *Eps* (including the point itself) – These are points that are the interior of a cluster

$\varepsilon = 1 \text{ unit}$, *MinPts* = 5

A **border point** has fewer than *MinPts* within *Eps*, but is in the neighborhood of a core point.

A **noise point** is any point that is not a core point nor a border point.

DBSCAN find clusters

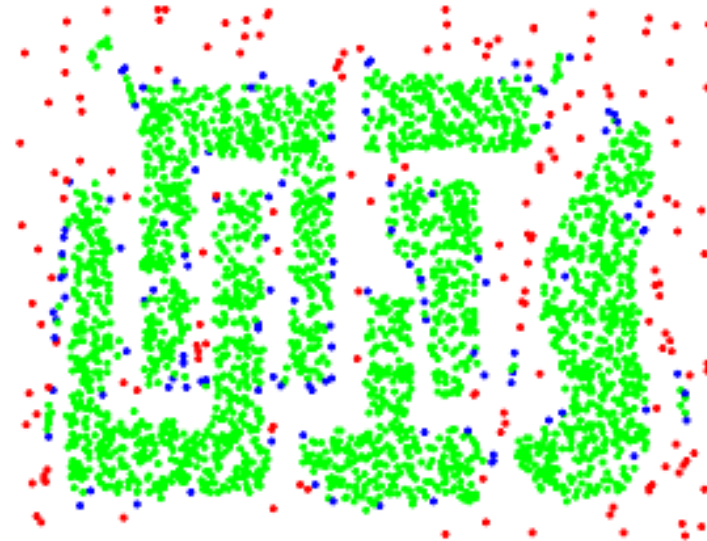


DBSCAN searches for clusters by checking the ϵ -neighborhood of each point in the database. If the ϵ -neighborhood of a point p **contains** more than $MinPts$, a new cluster with p as a core object is created. DBSCAN then iteratively collects directly density-reachable objects from these core objects, which may involve the merge of a few density-reachable clusters. The process terminates when no new point can be added to any cluster.

DBSCAN Example



Original Points



Point types: **core**,
border and **outliers**

$\epsilon = 10$, MinPts = 4

DBScan – Example (1)

- If Epsilon is 2 and minpoint is 2, what are the clusters that DBScan would discover with the following 8 examples:

$A1=(2,10)$, $A2=(2,5)$, $A3=(8,4)$, $A4=(5,8)$, $A5=(7,5)$,
 $A6=(6,4)$, $A7=(1,2)$, $A8=(4,9)$.

Matriks jarak (kuadrat):

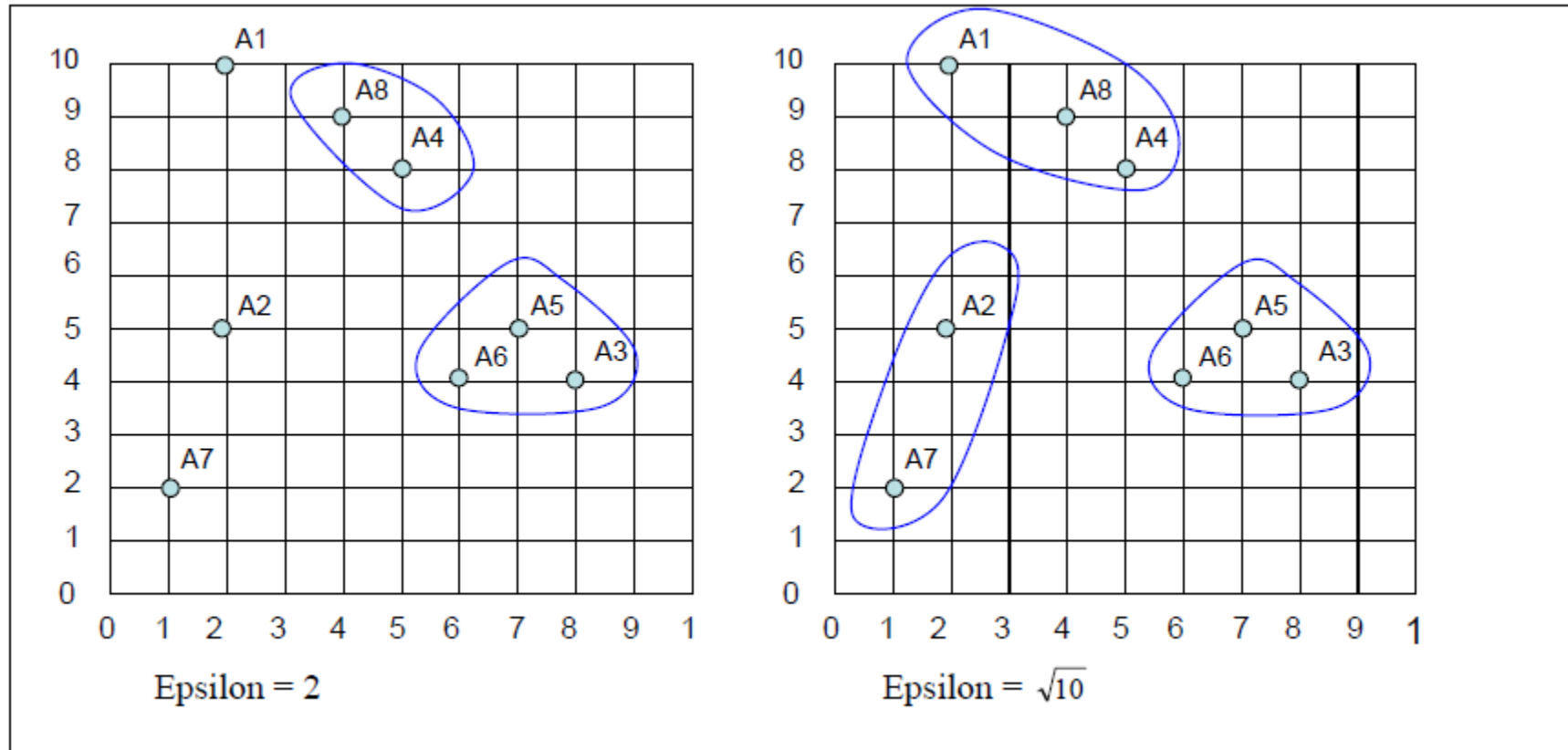
	A1	A2	A3	A4	A5	A6	A7	A8
A1	0	25	72	13	50	52	65	5
A2	25	0	37	18	25	17	10	20
A3	72	37	0	25	2	4	53	41
A4	13	18	25	0	13	17	52	2
A5	50	25	2	13	0	2	45	25
A6	52	17	4	17	2	0	29	29
A7	65	10	53	52	45	29	0	58
A8	5	20	41	2	25	29	58	0

DBScan – Example (2)

- Solutions:
- Epsilon neighborhood of each point
 - $N_2(A1)=\{\}$;
 - $N_2(A2)=\{\}$;
 - $N_2(A3)=\{A5, A6\}$;
 - $N_2(A4)=\{A8\}$;
 - $N_2(A5)=\{A3, A6\}$;
 - $N_2(A6)=\{A3, A5\}$;
 - $N_2(A7)=\{\}$;
 - $N_2(A8)=\{A4\}$
 - So A1, A2, and A7 are outliers, while we have two clusters $C1=\{A4, A8\}$ and $C2=\{A3, A5, A6\}$
- If Epsilon is $10^{1/2}$ then the neighborhood of some points will increase:
 - A1 would join the cluster C1 and A2 would join with A7 to form cluster $C3=\{A2, A7\}$.

	A1	A2	A3	A4	A5	A6	A7	A8
A1	0	25	72	13	50	52	65	5
A2	25	0	37	18	25	17	10	20
A3	72	37	0	25	2	4	53	41
A4	13	18	25	0	13	17	52	2
A5	50	25	2	13	0	2	45	25
A6	52	17	4	17	2	0	29	29
A7	65	10	53	52	45	29	0	58
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DBScan – Example (3)





04 Cluster Evaluation

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Outline

Cluster Validity

Internal Measures

External
Measures

Cluster Validity

- For supervised classification we have a variety of measures to evaluate how good our model is
 - Accuracy, precision, recall
- For cluster analysis, the analogous question is how to evaluate the “goodness” of the resulting clusters?
- But “clusters are in the eye of the beholder”!
- Then why do we want to evaluate them?
 - To avoid finding patterns in noise
 - To compare clustering algorithms
 - To compare two sets of clusters
 - To compare two clusters

Measures of Cluster Validity

- Numerical measures that are applied to judge various aspects of cluster validity, are classified into the following three types.
 - **Internal Index:** Used to measure the goodness of a clustering structure *without* respect to external information.
 - **Cohesion and Separation, Silhouette Coefficient**, Dunn Index, Cophenetic Correlation Coefficient (CPCC), etc.
 - **External Index:** Used to measure the extent to which cluster labels match externally supplied class labels.
 - Entropy, **Purity**, Precision, Recall, F-Measure, Rand Index, etc.
 - **Relative Index:** Used to compare two different clusterings or clusters.
 - Often an external or internal index is used for this function, e.g., SSE or entropy
- Sometimes these are referred to as **criteria** instead of **indices**
 - However, sometimes criterion is the general strategy and index is the numerical measure that implements the criterion.

Internal Measures: Cohesion and Separation

- **Cluster Cohesion:** Measures how closely related are objects in a cluster; compactness
 - Example: SSE
- **Cluster Separation:** Measure how distinct or well-separated a cluster is from other clusters; isolation
- Example: Squared Error
 - Cohesion is measured by the within cluster sum of squares (SSE)

$$WSS = \sum_i \sum_{x \in C_i} (x - m_i)^2$$

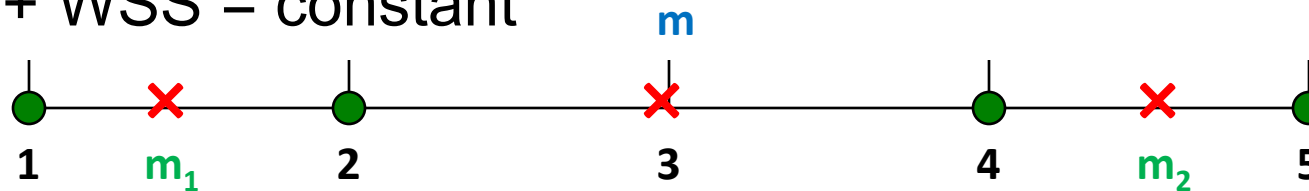
- Separation is measured by the between cluster sum of squares

$$BSS = \sum_i |C_i| (m - m_i)^2$$

- Where $|C_i|$ is the size of cluster i

Internal Measures: Cohesion and Separation

- Example: SSE
 - BSS + WSS = constant



K=1 cluster:

$$WSS = (1 - 3)^2 + (2 - 3)^2 + (4 - 3)^2 + (5 - 3)^2 = 10$$

$$BSS = 4 \times (3 - 3)^2 = 0$$

$$Total = 10 + 0 = 10$$

K=2 clusters:

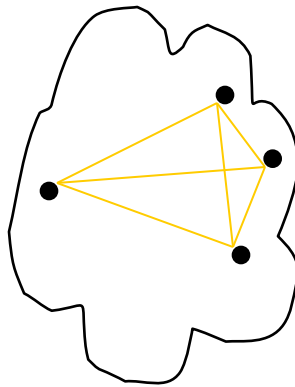
$$WSS = (1 - 1.5)^2 + (2 - 1.5)^2 + (4 - 4.5)^2 + (5 - 4.5)^2 = 1$$

$$BSS = 2 \times (3 - 1.5)^2 + 2 \times (4.5 - 3)^2 = 9$$

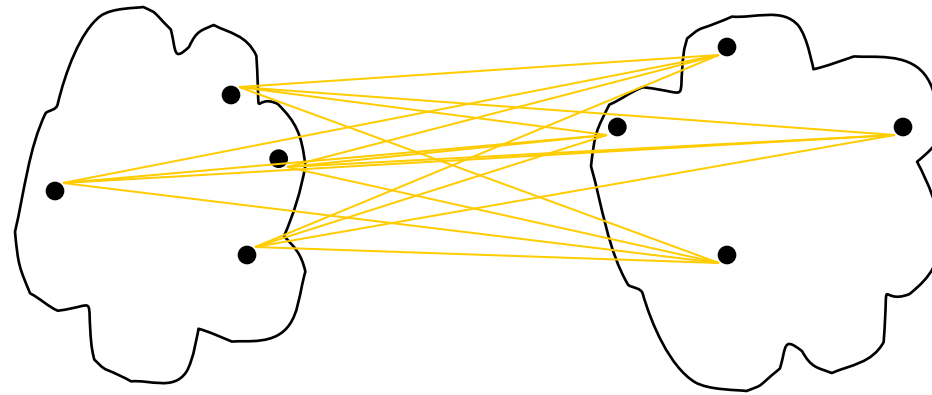
$$Total = 1 + 9 = 10$$

Internal Measures: Cohesion and Separation

- A proximity graph based approach can also be used for cohesion and separation.
 - Cluster cohesion is the sum of the weight of all links within a cluster.
 - Cluster separation is the sum of the weights between nodes in the cluster and nodes outside the cluster.



cohesion

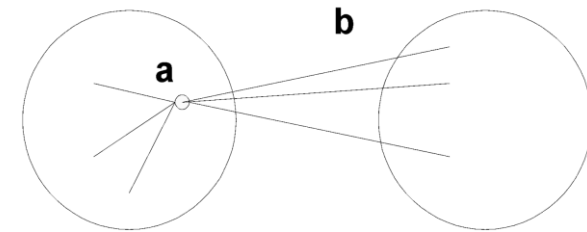


separation

Internal Measures: Silhouette Coefficient

- Silhouette Coefficient combine ideas of both cohesion and separation, but for individual points, as well as clusters and clusterings.
- For an individual point, i
 - Calculate a = average distance of i to the points in its cluster
 - Calculate b = min (average distance of i to points in another cluster)
 - The silhouette coefficient for a point is then given by

$$s = \frac{b - a}{\max(a, b)}$$



- The score is bounded between -1 for incorrect clustering and +1 for highly dense clustering.
 - Scores around zero indicate overlapping clusters.
- Can calculate the Average Silhouette width for a cluster or a clustering

Silhouette Coefficient (SC) Example

$$s = \frac{b - a}{\max(a, b)}$$

Cluster 1:

Point A1: (2, 5)

Point A2: (3, 4)

Point A3: (4, 6)

Cluster 2:

Point B1: (8, 3)

Point B2: (9, 2)

Point B3: (10, 5)

Cluster 3:

Point C1: (6, 10)

Point C2: (7, 8)

Point C3: (8, 9)

Distance:

SC A1:
Step 1:

- Distance from Point A1 to Point A2: $\sqrt{(2-3)^2 + (5-4)^2} = \sqrt{2}$
- Distance from Point A1 to Point A3: $\sqrt{(2-4)^2 + (5-6)^2} = \sqrt{5}$
- Average cohesion for Point A1 : **$a = (\sqrt{2} + \sqrt{5}) / 2 \approx 1.825$**

Step 2a:

- Distance from Point A1 to Point B1: $\sqrt{(2-8)^2 + (5-3)^2} = \sqrt{40}$
- Distance from Point A1 to Point B2: $\sqrt{(2-7)^2 + (5-8)^2} = \sqrt{34}$
- Distance from Point A1 to Point B3: $\sqrt{(2-8)^2 + (5-9)^2} = \sqrt{52}$
- Average separation for Point A1 to cluster 2 = $(\sqrt{40} + \sqrt{34} + \sqrt{52}) / 3 \approx 6.456$

Step 2b:

- Distance from Point A1 to Point C1: $\sqrt{(2-6)^2 + (5-10)^2} = \sqrt{26}$
- Distance from Point A1 to Point C2: $\sqrt{(2-7)^2 + (5-8)^2} = \sqrt{18}$
- Distance from Point A1 to Point C3: $\sqrt{(2-8)^2 + (5-9)^2} = \sqrt{32}$
- Average separation for Point A1 to cluster 3 = $(\sqrt{26} + \sqrt{18} + \sqrt{32}) / 3 \approx 5.333$

Minimum separation A1 to other cluster: **$b = \min(6.456, 5.333) = 5.333$**

Step 3:

SC A1 = $(5.333 - 1.825) / \max(5.333, 1.825) \approx 0.657$

- Count SC for all points
- SC for a cluster is the average SC points in the cluster
- SC for all cluster is the average SC of all clusters/ all points

External Measures: Purity

- Purity: ratio between the dominant class in the cluster and the size of the cluster.

$$Purity(w_i) = \frac{1}{n_i} \max_j (n_{ij}) \quad j \in C$$

- Assume documents with C gold standard classes, while our clustering algorithm produce K clusters, $w_1, w_2, \dots, w_k$ with n_i members.

Clustering for 'Labeled Data'

Purity

Bad clustering:
Purity close to 0

Good clustering:
Purity close to 1

Purity in a
cluster:

$$P(C_r) = \frac{1}{n_r} \max_i n_r^i$$

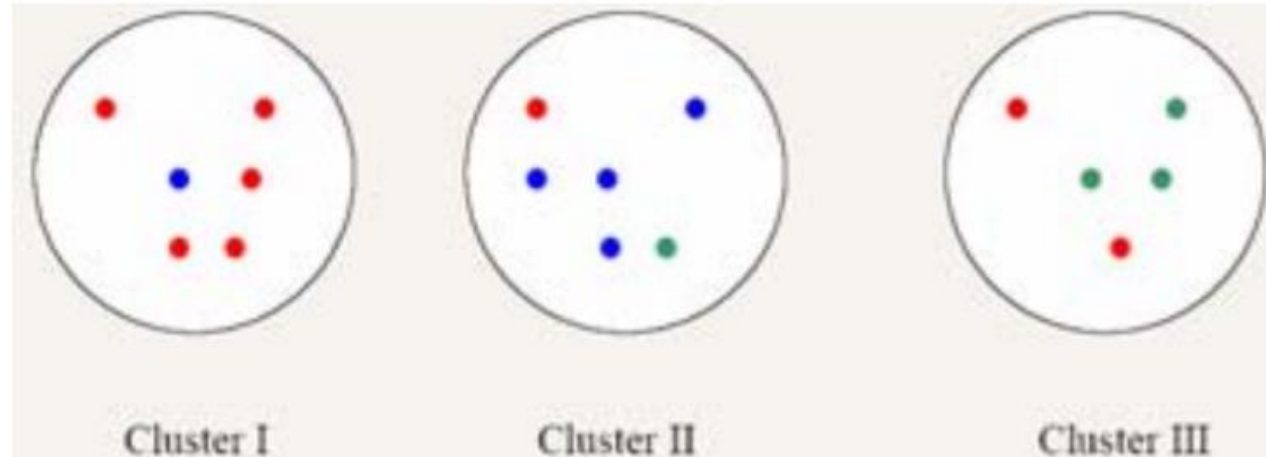
Purity of all clustering
result:

$$\begin{aligned} Purity(C) &= \sum_{r=1}^k \frac{n_r}{n} P(C_r) \\ \text{or} &= \frac{1}{n} \sum_{r=1}^k \max_i (n_r^i) \end{aligned}$$



External Measures: Purity

- Example:



- Cluster I: Purity = $1/6 (\max(5, 1, 0)) = 5/6$
- Cluster II: Purity = $1/6 (\max(1, 4, 1)) = 4/6$
- Cluster III: Purity = $1/5 (\max(2, 0, 3)) = 3/5$
- All cluster: Purity = $1/17 (5+4+3) = 12/17$

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